Blending Panoramas

Linear Blending

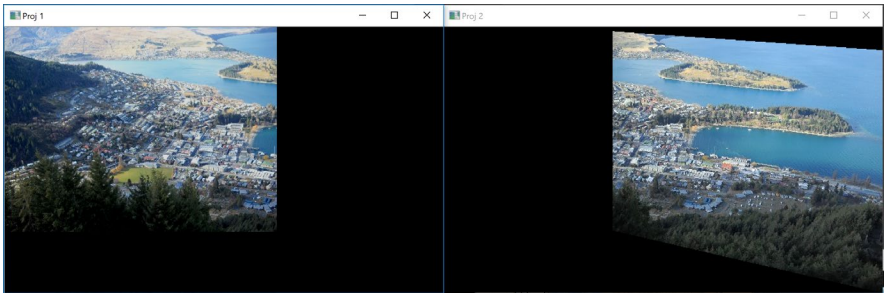
Get average color difference between overlapping images and apply global change to one of the images to compensate.

- Compute difference between RGB values for each overlapping pixel in stitched image:
  - image2[i,j,k] – image1[i,j,k]
- Get average of all these differences
- Add average RGB difference to each pixel in image1

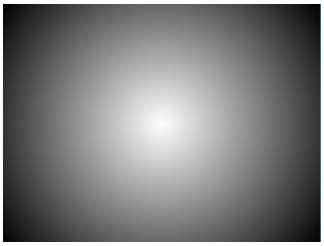
Multi-Band Blending

Break image into high and low (more blurred) frequency components. Blend high frequency areas over shorter ranges and low frequency areas over longer ranges.

- Create two images the size of your panorama. In each image only map over one of the original images showing where it should lay in the stitched image. These will be referred to as the *mapped color* images.



- Compute *spherical* weights for each image. The center pixel gets a weight of 1 and the corners get a weight of 0. Every other pixel gets a weight between [0-1] based on its distance from the center of the image.
  - Compute the length of the diagonal line connecting the center of the image to one corner (diag).
  - Compute the distance between a pixel and the center of the image (d).
  - Pixel's weight = 1 – d/diag



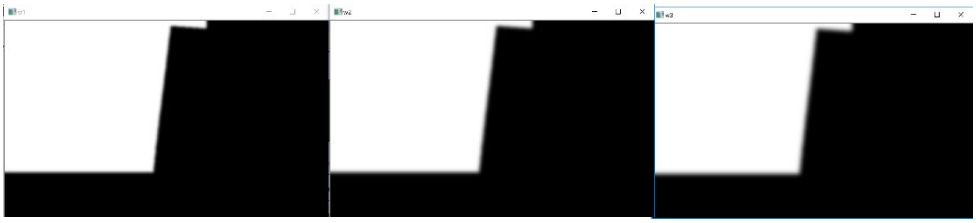
- Compute *blending* weights for each image. Iterate through the pixels in image1 (x1, y1) and map them to image2 (x2, y2). If the spherical weight stored at (x1, y1) is higher than the spherical weight at (x2, y2) then (x1, y1) gets a blending weight of 1. Otherwise its blending weight is 0. If (x1, y1) does not have a correspondence in image2 (it's outside image2's bounds), give it a weight of 1. Repeat this process for image2, mapping all of its pixels to image1 and comparing.

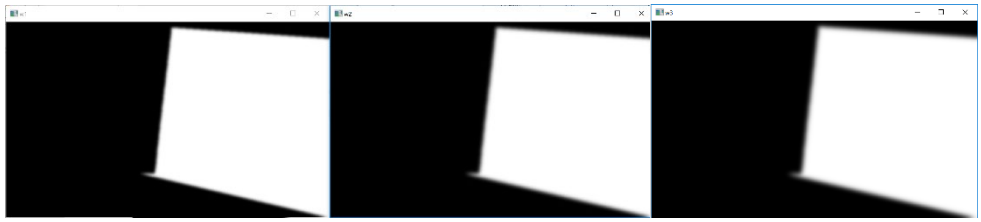


- Map the blending weights to the stitched image. Keep a separate image/variable for each image's blending weights.

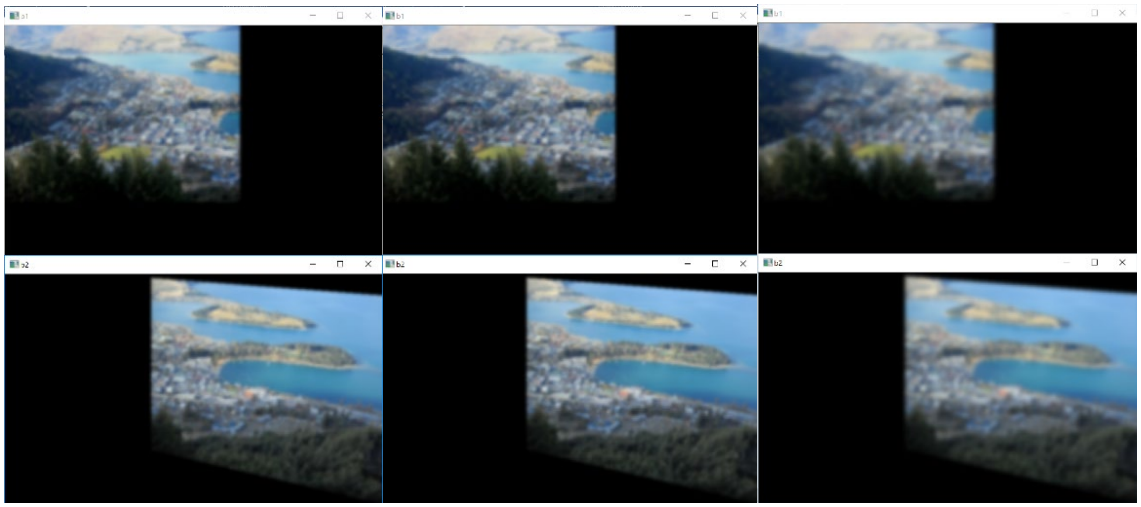


- Blur the blending weights images three times each by convolving with a Gaussian filter.
    - For each image, this will give you three levels (bw1, bw2, bw3)

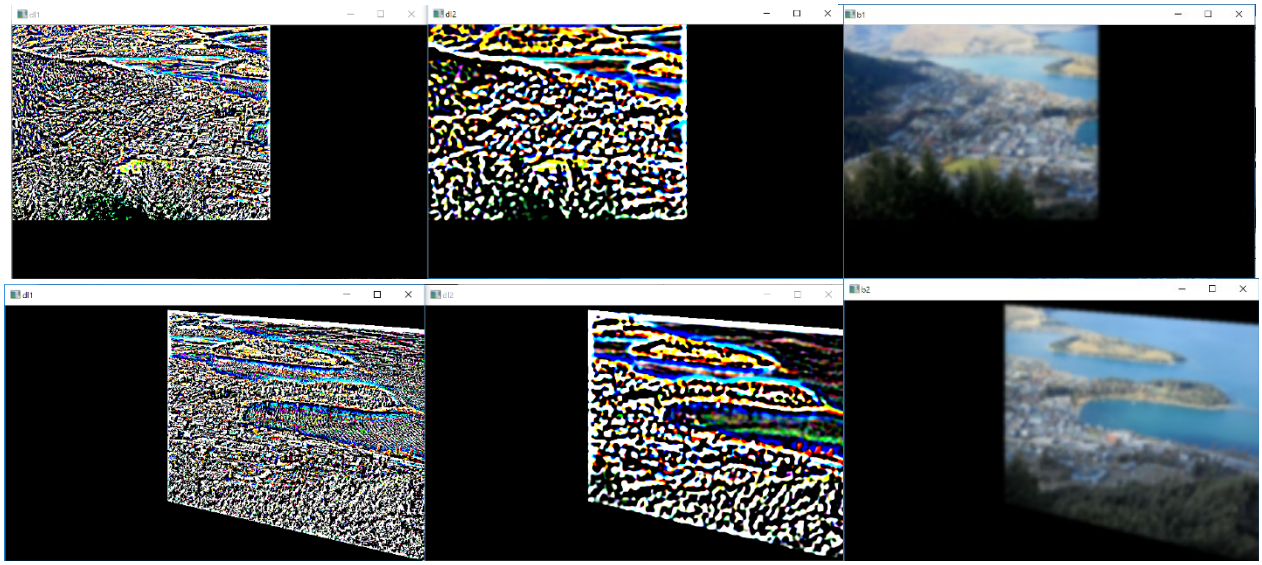




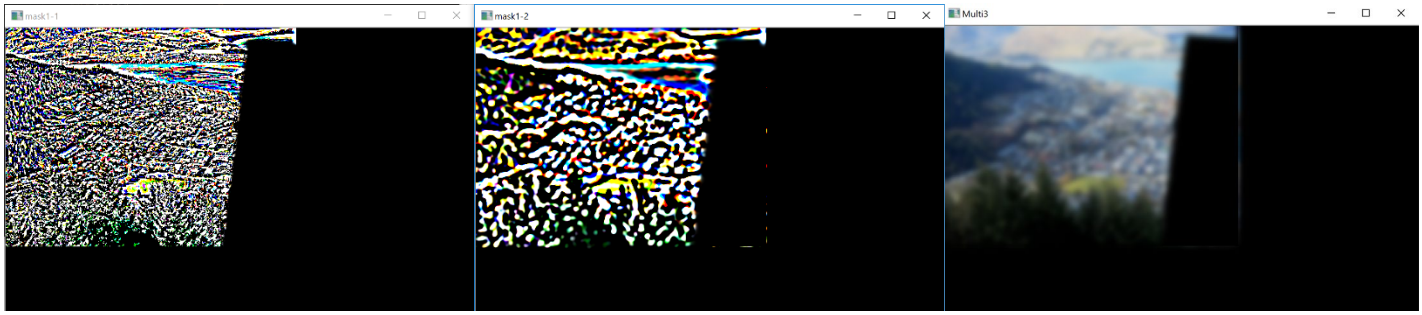
- Blur the mapped color images three times each by convolving with a Gaussian filter.
    - For each image, this will give you three levels (blur1, blur2, blur3)

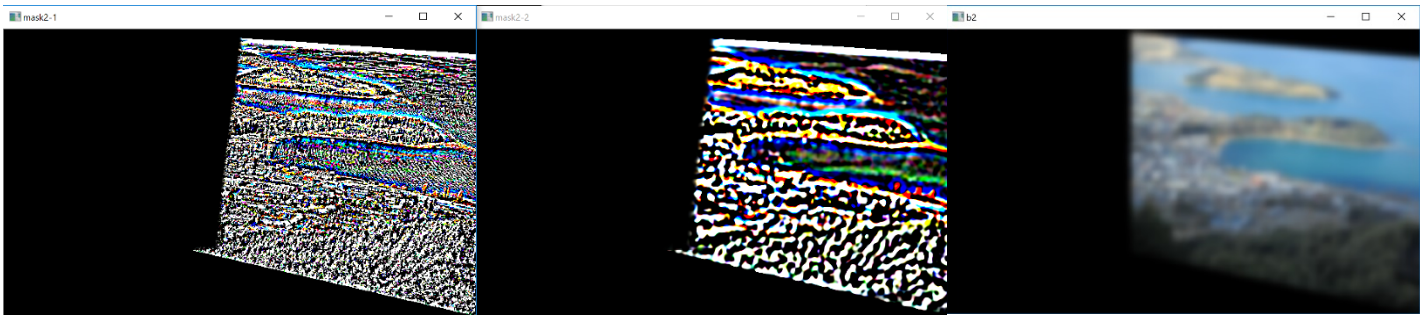


- Get the two difference of Gaussians for each image by subtracting the first blurred image from the mapped color image and the second blurred image from the first blurred image.
    - Combining these with the most blurred color mapped image, for each image, this will give you three levels (dg1, dg2, blur3)



- Combine the corresponding levels of the difference of Gaussian images and the blurred blending weights images pixel-wise.
    - For each image, this will give you three levels (c1 = dg1*bw1, c2 = dg2*bw2, c3 = blur3*bw3)





- Reconstruct each level of the panorama by combining together the two images being stitched together. Add corresponding pixels together and divide by the sum of weights of those pixels
    - r1[i,j] = (c1_im1[i,j] + c1_im2[i,j])/(bw1_im1[i,j]+bw1_im2[i,j])
    - r2[i,j] = (c2_im1[i,j] + c2_im2[i,j])/(bw2_im1[i,j]+bw2_im2[i,j])
    - r3[i,j] = (c3_im1[i,j] + c3_im2[i,j])/(bw3_im1[i,j]+bw3_im2[i,j])
- Add the weighted levels together to get your final blended stitched image
    - finalReconstructed = r1 + r2 + r3

